CVA-30 Series Pneumatic Control Valve Actuator

Model Series: CVA-30 | Equipment Type: Pneumatic Spring-Diaphragm Actuator

1. Overview

The CVA-30 Series is a spring-opposed, diaphragm-type pneumatic actuator used to position globe and control valves in modulating service across BRIMIS process lines. It is typically paired with a positioner for precise flow control and fails to a pre-set safe position (fail-open or fail-closed, order-dependent) on loss of air supply.

2. Technical Specifications

Parameter	Value
Actuator sizes	CVA-30-100 / -200 / -400 (diaphragm area, cm²)
Supply air pressure	1.4 – 7.0 bar
Fail-safe action	Fail-closed (spring-to-close) or fail-open (spring-to-open)
Spring range	0.2 – 1.0 bar (standard), 0.4 – 2.0 bar (extended)
Positioner	CVA-POS electro-pneumatic, 4–20mA input
Enclosure rating	IP66 / NEMA 4X
Ambient temperature range	-20°C to +80°C
Stem travel	20 – 100mm depending on actuator size

3. Installation Guidelines

1. Verify fail-safe action on the nameplate matches the process safety requirement before installation — this cannot be field-reversed without a spring change kit.
2. Mount actuator to valve yoke per the torque values in Section 4; do not use an impact wrench on yoke nuts.
3. Connect instrument air supply through a filter-regulator set to the actuator's rated supply pressure; verify with a calibrated gauge.
4. Calibrate the positioner zero and span against the valve's actual open/closed travel limits, not the actuator's full mechanical travel.
5. Stroke-test the valve from the control system at 0%, 50%, and 100% demand and confirm position feedback agrees within ±2%.

4. Torque Specifications

Apply torque values below using a calibrated torque wrench. Over-torquing bolted connections can distort sealing faces and cause premature gasket or packing failure.

Fastener / Location	Size	Torque (N·m)
Yoke-to-actuator mounting bolts	M12	55
Yoke-to-actuator mounting bolts (CVA-30-400)	M16	95
Stem connector clamp bolt	M10	35
Positioner mounting bracket bolts	M8	15
Diaphragm casing bolts	M10	30

5. Maintenance Schedule

Interval	Task
Monthly	Check instrument air filter-regulator for condensate accumulation and drain if needed.
Quarterly	Stroke-test full travel and verify stroking time against commissioning baseline; a slowing stroke time oft
Annually	Inspect diaphragm for cracking at the clamped edge; replace preventively on actuators in severe-duty c
Annually	Verify fail-safe action by isolating air supply and confirming the valve drives fully to its safe position with
Every 5 years	Replace actuator spring set and diaphragm as a precaution regardless of observed condition in safety-

6. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Valve does not reach full travel	Supply air pressure below spring range	Verify supply pressure at actuator; recalibrate
Slow or erratic stroking	Diaphragm beginning to crack, or air filter-regulator restricted	Inspect diaphragm; clean or replace filter-regulator element
Valve fails to a position slower than the specified time	Partially blocked quick-exhaust valve, or degraded spring	Inspect and clean quick-exhaust valve; if time still fails spec
Positioner output unstable / hunting	Positioner tuning too aggressive for valve friction	Re-tune positioner gain; inspect feedback linkage for wear/
Air leakage audible at actuator body	Diaphragm rupture or O-ring seal failure	Isolate air supply immediately; replace diaphragm/casing se

7. Safety Warnings

■ Never disassemble an actuator under spring tension without first following the OEM spring-compression procedure — stored spring energy can cause serious injury.

■ Confirm fail-safe action before every installation; an incorrectly specified fail-action actuator can defeat a process safety interlock.

■ Isolate instrument air supply and bleed residual pressure before any maintenance.

■ Do not exceed the maximum supply pressure stamped on the nameplate — over-pressurizing the diaphragm casing can cause it to rupture violently.